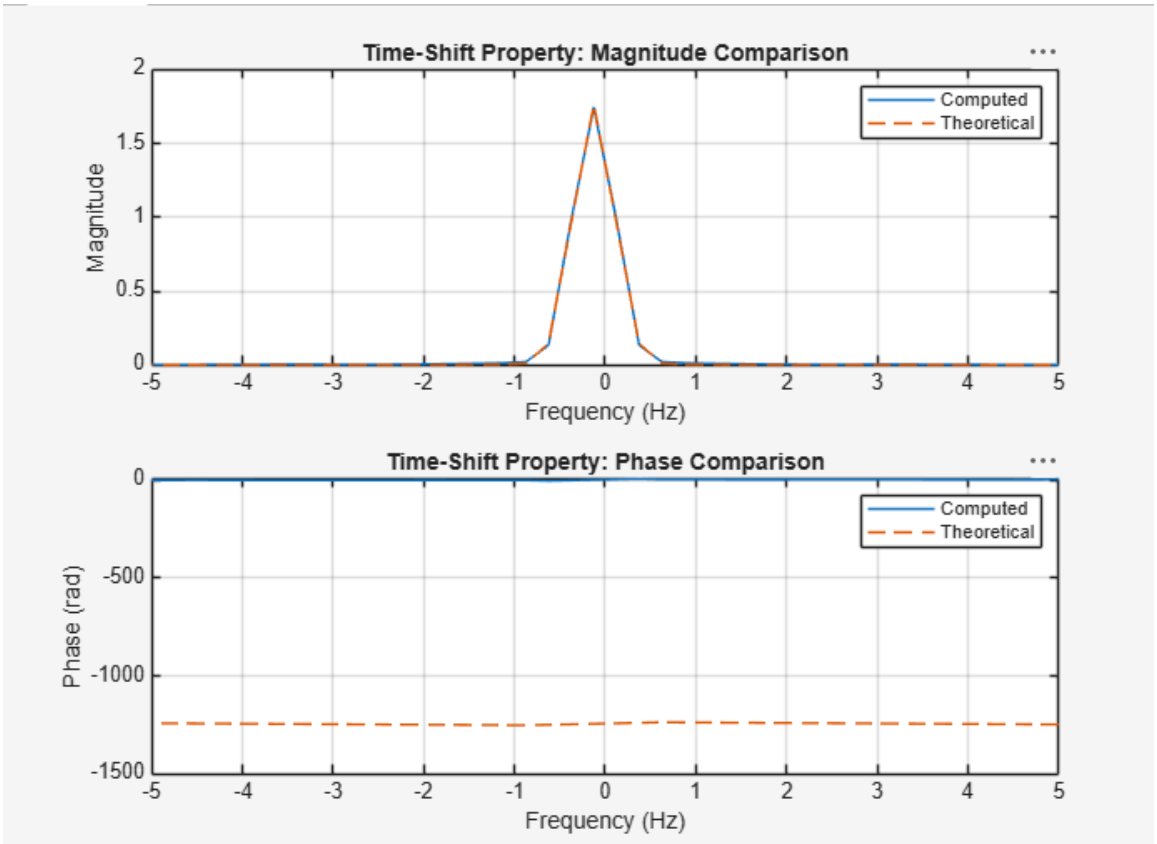
Experiment – 7:
Continuous-Time Fourier Transform and Spectrum

(iii)

/MATLAB Drive/Experiment_7_4.m

```
clc; clear; close all;
%% 7(iii ) Verification of Fourier Transform Properties and Frequency-Domain
figure;
% Verification of Fourier Transform Properties
t = -2:0.001:2;
dt = t(2)-t(1);
x = exp(-t.^2);
x_shift = interp1(t,x,t-0.4,'linear',0);
X = fftshift(fft(x))*dt;
N = length(x);
f = (-N/2:N/2-1)/(N*dt);
% Time-shift property: x(t-t0) <-> X(f)e^(-j2*pi*f*t0)
X_shift_theory = X.*exp(-1j*2*pi*f*0.4);
X_shift = fftshift(fft(x_shift))*dt;
subplot(2,1,1);
plot(f,abs(X_shift),f,abs(X_shift_theory),'--');
xlim([-5 5]); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude');
title('Time-Shift Property: Magnitude Comparison');
legend('Computed','Theoretical');
subplot(2,1,2);
plot(f,unwrap(angle(X_shift)),f,unwrap(angle(X_shift_theory)),'--');
xlim([-5 5]); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
title('Time-Shift Property: Phase Comparison');
legend('Computed','Theoretical');
```

Code



Output